Telco AI Operations Assistant

A production-style multi-agent AI assistant for support and NOC teams: it answers from your documentation with citations, investigates customers and incidents through secure tools, and changes systems only after human approval.

▶ Live demo: **telco-ai-ops.vercel.app** ⌘ Code: **github.com/weblinkify/llm-agent-platform**

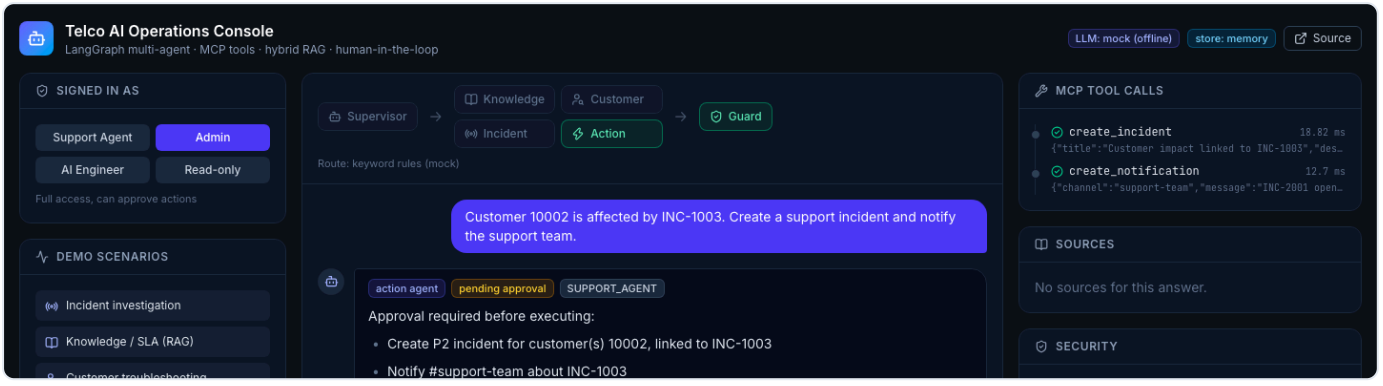
WHAT IT DOES FOR YOUR BUSINESS

- **Reduces L1 load:** cited answers from SLAs and runbooks, plus customer diagnoses with next steps.
- **Safe AI actions:** the AI proposes; a different authorised person approves; only then do tools run.
- **Measurable quality:** a 48-case eval suite gates every release.
- **Enterprise security:** JWT auth, role-based tool access, injection defence, PII masking, audit log.
- **Runs on your cloud:** Docker + Terraform for Azure; Azure OpenAI, OpenAI or Anthropic.

MEASURED, NOT CLAIMED

Routing accuracy	**100%**
Retrieval recall@4	**95.8%**
Tool-call correctness (incl. approvals)	**100%**
Prompt-injection attacks blocked	**6 / 6**
Automated tests / coverage	**93 / 93%**
Browser end-to-end tests	**7 / 7**

Mock-mode eval suite run in CI on every change; the LLM is a deterministic test double, so these figures measure retrieval, tools, approvals and security.



DELIVERED IN THIS PROJECT

- LangGraph multi-agent workflow with pause/resume approvals (Postgres checkpoints)
- MCP server with 12 typed tools; works with Claude Desktop
- Hybrid RAG: BM25 + pgvector, rank fusion, reranker, "insufficient context" guard
- Next.js console: streaming, agent trace, approval inbox, role switcher
- OpenTelemetry tracing, Prometheus metrics, cost per request
- CI/CD: tests, evals, e2e, security scans, container scan, Terraform

TECHNOLOGY

Python 3.12 · FastAPI · LangGraph · MCP SDK · Pydantic · PostgreSQL + pgvector · Azure OpenAI · OpenAI · Anthropic · Next.js 16 · TypeScript · Tailwind · Playwright · Docker · Terraform · Azure Container Apps · Key Vault · OpenTelemetry · Jaeger · Prometheus · GitHub Actions

ADAPT IT TO YOU

Swap the synthetic data for your ticketing, CRM and docs (Jira, ServiceNow, Salesforce, SharePoint) by adding MCP tools: the agents, approvals, security and evals carry over.

Run it yourself in one command: **docker compose up** (no API keys needed)

All data is synthetic